

HIGH LEVEL, AB, Canada

WMO: 710660

Lat: 58.620N

Lon: 117.160W

Elev: 338

StdP: 97.33

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.6	(c) -35.2	(d) -43.1	(e) 0.1	(f) -38.5	(g) -40.2	(h) 0.1	(i) -35.1	(j) 8.5	(k) -13.6	(l) 8.0	(m) -15.3	(n) 0.2	(o) 0	(p) 0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.3	(c) 27.7	(d) 17.2	(e) 26.0	(f) 16.5	(g) 24.3	(h) 15.8	(i) 18.7	(j) 25.2	(k) 17.7	(l) 24.1	(m) 16.7	(n) 22.7	(o) 3.1	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.0	21.3	15.1	11.2	20.2	14.2	10.5	19.3	54.3	25.3	51.2	24.2	48.3	22.8	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.8	6.0	-41.6	30.9	3.5	1.6	-44.1	32.0	-46.2	33.0	-48.1	33.9	-50.6	35.0
			-41.5	20.5	3.1	1.2	-43.7	21.4	-45.5	22.1	-47.2	22.8	-49.5	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-19.6	-16.1	-9.4	1.7	9.5	14.6	16.9	14.8	9.2	1.2	-10.9	-17.7		
	DBStd	14.42	8.61	7.46	8.37	5.96	4.71	3.34	2.74	3.31	4.13	4.62	6.85	7.88		
	HDD10.0	4399	919	731	602	251	64	5	0	4	62	275	626	860		
	HDD18.3	6869	1177	964	860	499	274	117	60	117	274	532	876	1118		
	CDD10.0	601	0	0	0	3	49	144	214	152	38	2	0	0		
	CDD18.3	29	0	0	0	0	1	6	16	7	0	0	0	0		
	CDH23.3	571	0	0	0	1	41	131	251	132	14	1	0	0		
	CDH26.7	97	0	0	0	0	6	19	49	23	1	0	0	0		
	WSAvg	2.3	1.9	2.2	2.7	3.0	2.9	2.5	2.1	2.0	2.2	2.3	2.0	1.7		
(15) Precipitation	PrecAvg	441	26	19	22	18	43	67	62	57	36	34	29	24		
	PrecMax	644	41	39	60	56	86	135	107	125	90	100	56	47		
	PrecMin	337	6	2	3	5	14	16	13	2	9	1	0	10		
	PrecStd	75	11	11	16	12	19	30	25	31	22	26	17	11		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	5.5	11.0	21.1	27.3	28.9	30.2	29.4	25.5	18.3	7.2	5.6		
		MCWB	1.0	1.5	4.6	10.3	14.6	17.0	18.8	18.4	16.2	10.8	2.9	2.2		
	2%	DB	-0.3	1.4	7.8	16.8	24.1	26.6	28.0	26.6	22.3	13.6	3.1	0.2		
		MCWB	-2.1	-1.2	2.6	8.0	12.8	16.0	17.7	17.5	14.3	7.9	0.6	-1.6		
	5%	DB	-4.4	-2.3	5.2	14.0	21.7	24.7	26.1	24.6	19.6	10.9	0.5	-4.5		
		MCWB	-5.2	-4.1	1.2	6.1	11.6	15.2	17.2	16.7	12.7	6.1	-0.7	-5.3		
	10%	DB	-8.4	-5.5	2.7	11.6	19.3	22.8	24.3	22.6	17.4	8.5	-1.6	-7.2		
		MCWB	-9.0	-6.6	-0.4	5.0	10.4	14.4	16.4	15.6	11.4	5.0	-2.5	-7.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.6	4.9	10.8	15.7	18.9	20.4	20.0	17.0	11.9	3.2	2.4		
		MCDB	4.3	5.0	10.3	20.5	24.5	25.6	26.5	26.8	24.3	17.3	6.3	5.7		
	2%	WB	-2.0	-1.2	2.8	8.3	13.7	17.2	18.8	18.4	15.0	8.1	0.9	-1.6		
		MCDB	-0.6	1.4	7.3	16.1	22.2	24.0	25.7	24.7	21.0	12.5	2.5	0.0		
	5%	WB	-5.3	-4.0	1.3	6.5	12.2	16.1	18.0	17.3	13.4	6.6	-0.6	-5.2		
		MCDB	-4.3	-2.3	5.1	13.1	20.1	22.8	24.6	23.3	18.3	10.3	0.6	-4.5		
	10%	WB	-9.0	-6.6	-0.4	5.2	10.9	15.1	17.1	16.2	12.0	5.1	-2.5	-7.8		
		MCDB	-8.4	-5.5	2.6	11.1	18.3	21.3	23.1	21.4	16.3	8.2	-1.6	-7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	11.3	12.5	12.3	13.8	12.8	12.3	12.6	12.3	8.9	8.2	8.9		
		MCDBR	14.1	14.8	14.3	16.6	18.0	16.6	16.1	16.2	16.9	14.3	9.7	12.3		
		MCWBR	12.4	12.5	10.0	9.5	8.8	7.8	7.4	8.2	9.6	9.2	7.9	10.8		
		MCDBR	13.9	14.6	14.0	15.9	16.7	14.1	14.5	14.7	15.7	13.3	9.2	12.0		
	5% WB	MCWBR	12.3	12.4	10.0	9.4	8.4	7.1	7.2	7.9	9.3	8.9	7.6	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.215	0.256	0.272	0.333	0.344	0.362	0.369	0.350	0.311	0.283	0.240	0.198		
	taud		2.241	2.276	2.290	2.314	2.423	2.401	2.398	2.443	2.554	2.561	2.284	2.112		
	Ebn at Noon		665	792	883	874	889	873	858	850	841	770	613	539		
	Edh at Noon		42	68	91	107	103	107	105	94	72	53	40	33		
(44) All-Sky Solar Radiation	RadAvg		0.55	1.48	3.16	4.89	5.72	6.00	5.64	4.57	3.09	1.49	0.58	0.33		
	RadStd		0.04	0.10	0.19	0.22	0.42	0.46	0.26	0.30	0.23	0.18	0.07	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+1.90	+2.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	+1.59	+2.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page